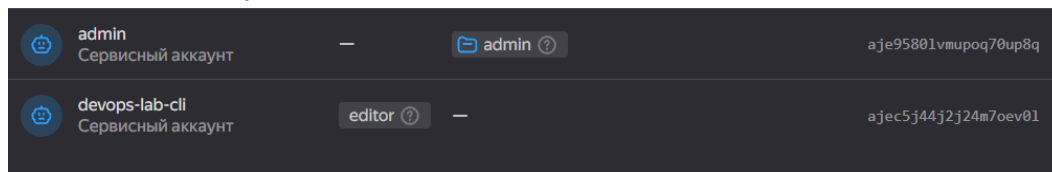


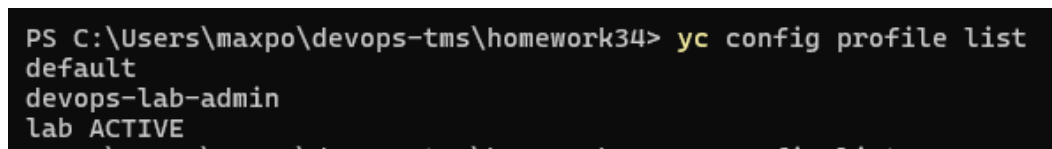
ДЗ выполнено в Яндекс Cloud.

Предусловия:

Создано два аккаунта: admin и devops-lab-cli.



А также два профиля:



devops-lab-admin:

```
service-account-key:
  id: aje6aeb8tde9ded2dnhh
  service_account_id: aje95801vmupoq70up8q
  created_at: "2026-08-25T20:30:21.877568841Z"
  key_algorithm: RSA_2048
  public_key: |
    -----BEGIN PUBLIC KEY-----
    MIIETjANBgkqhkiG9w0BAQEFAAOCQAQ8AMIIBCgKCAQEAzbiNCzIuqQGH6Fa/pjv
    /up/Pg9THfM9I3A1aYLj083Fuivt5H8G3TigeBKao1zgshIRzNsPzNr9HhZz8P3
    DRqSDuFKDc7+wQ0daBNaRScPyNB9YH/6yBiwJIPVlnpi8eAxUk117Cxcni19uKzY
    YXZdShkMrwzrDLYsGEoB2H27w/oxmZw7BV2+mnNhbtXVAXDBjs7/o9TDhnX09Hyw
    u0iul1U5ZKC/dff2UnOvSVmDFsGLsr/7GJGs/5FyhUTETUYzLB5cxVBbx0SAzGkL
    N2evca2UNFC006ce5WBMiEuCHJDrtWTWG2Y2V5maQwAbDw21Q4iP+Vy0T00x6sPYJ
    pQIDAQAB
    -----END PUBLIC KEY-----
  private_key: |
    PLEASE DO NOT REMOVE THIS LINE! Yandex.Cloud SA Key ID <aje6aeb8tde9ded2dnhh>
    -----BEGIN PRIVATE KEY-----
    MIIETjANBgkqhkiG9w0BAQEFAAOCQAQ8AMIIBCgKCAQEAzbiNCzIuqQGH6Fa/pjv
    /up/Pg9THfM9I3A1aYLj083Fuivt5H8G3TigeBKao1zgshIRzNsPzNr9HhZz8P3
    DRqSDuFKDc7+wQ0daBNaRScPyNB9YH/6yBiwJIPVlnpi8eAxUk117Cxcni19uKzY
    YXZdShkMrwzrDLYsGEoB2H27w/oxmZw7BV2+mnNhbtXVAXDBjs7/o9TDhnX09Hyw
    u0iul1U5ZKC/dff2UnOvSVmDFsGLsr/7GJGs/5FyhUTETUYzLB5cxVBbx0SAzGkL
    N2evca2UNFC006ce5WBMiEuCHJDrtWTWG2Y2V5maQwAbDw21Q4iP+Vy0T00x6sPYJ
    pQIDAQAB
    -----END PRIVATE KEY-----
cloud-id: blgt9shupim2k9j5gomg
folder-id: blq6gpp7fmeo0hlca4ue
```

lab:

```
service-account-key:
id: aje1rfla4i4pmrv5l78
service_account_id: aje5j4uj2j24m7oev0l
created_at: "2026-08-25T21:38:23.433369840Z"
key_algorithm: RSA_2048
public_key: |
-----BEGIN PUBLIC KEY-----
MII8IjANBgkqhkiG9w0BAQFAAOCAQEAQ8AMIIBgKCAQEAq1hire815gbc1fPtg6e/
kS/GFwo1UFZY0zXvK3TY0HgkMTQFEjLWDVdDozs1hXcwHvaeVa8Ze9cqcD0u3Tj7
72CS2L6EVcaDUxpDLuxUTq+vXyYOR/N9R9tN17KEUHLZPa3hACAL9Lj3S0e6mU1
tL4hqRA+eJk/LdgDnbFo+gxebYRBuhZBzc0wkciv5IfwLE13vphAQ/XHybepnwBn
QIQVyn270kasadb72cH7MVPvRgTIE0doU4VZUJgsREzLr/PXDVIDXWNX4/zjaQH
ewWoAS2oIlaEaaEv7m3dZ37tOuJzqW00RMS8TKhXsesWA24TxuJ1NmCAKTN7UOL
EWIDAQAB
-----END PUBLIC KEY-----
private_key: |
PLEASE DO NOT REMOVE THIS LINE! Yandex.Cloud SA Key ID <aje1rfla4i4pmrv5l78>
-----BEGIN PRIVATE KEY-----
MII8IjANBgkqhkiG9w0BAQFAAOCAQEAASCBKwggSIAgEAAoIBAQCmWgkT7zXmBtyV
8+2Dp7+RL8YVajvQVLjTNe8rdNg4eAoxOoU5OVYNV00jOzWfDzAe9p5Vrx171ypw
PS7d0PvvYJLYvoRVxoNTGkMu7FR0r69fJg5H831H202LsoRQevK9reEAICUD0uPd
Kh7qZTW0v1GpED54mT+V2A0dsWj6DF5thEG6FkHJzTCRyK/kh/CUTXe+mEBD9cfJ
t6mfAgdCBXKfbvSRQoMHvZwFxsUim9GBMgQ52hThVlQmCxEtOwv89cNUGNdY1f
j/OMBAd7Bag8Lag1VoRpoS/vDd1nFu066NmparTRExLmQFex6xYBnhPG4kG22wA
pM3tQ6UTAgMBAAECggEAEzk7j0OCJXbnepmvCuAN0QwyVVPziHQ0hbbyTxV5Z2
3byZXdW5Gd0W9auL3pOU4044aTJK5LYupDZjyhZRW0go0mpkKseikD8BTb5pRe
7eqjRoGFQqhb+RDWVLQ4kX5+H+StWc0a2+A8RGTL7dye8n8+3LsFzgZiucOQVHV
yStD82UHKV/S4m1kggMaGPQj/cC8659Dj8NzknnVogrWLNqBRUUsNvVfp9uwpD
pWu98enSntbJkt1RFVnZknVnFKH/KU/a7Xeb/zGqioT6J06FF80Q0BF1FqU4e
9BhSE3Jrtjygd38CgUwM6dWpmQ+/j1H546eKFIpQIQH8gQDmkiZ82y639ivTAXS
ax1AeYHcYV3WvOePHzXGyrYeJavkIPStAurR+C+9jWuee6681Nwp7y/Lq3z+gWp
vzaBTWq62K9j/iS84deyVA/sk1GRvQ36lQ3aNNCq15jA2yucNaps/z8U3lgt8B
mPLFCLAgTD5JyRjmOdjY20iQK8QDwa+NFM33XcuL726Q5256613669LdVaPq9
6baF20kj+tdEFV4VAPWd64pzcZkmNzzsT6OL/deOJZtH7qHr+sW5N1VhMgkM4h1
ksgSnzE7MEq/Lrrnxw7aZiYb/3DfUabAdGgiyaRRWCacv9+nw/OP/GFu6Cmaaeyz
W5jelhNduwKBgCYbt9xGwkmkt0MqLbvqjYyOoPUemZltsY+T5LgOfpyL534i4s
9tImQLXxY8+M5179iHsF5w4UgX23ADzyH5QcGQDAKs2WxBmXnUSgCzSYdZbCiB
BkQ5xxRAUS9M/2R9XGHmmXE8XCI5UJZKlqvFuK46RdNnNimLM4nNte9hAogAPJsq
FFL07I1c88Crrp7g8PKRI47eHLQbi2pv6M6yULTB+88+YzLUm0DMVn0aek7MTY8F
DyckuMzwYxiUM/shvFe3jyaEsgz02UyvuH0+mpg/yGIQ+KpNbHVYryEztppav3G
kc6mf0ePbBrFwk+mPGDzTsJN9ZNTec8numJ4/tKcgyBrMRomLpQyFuZEszLUQv+
QcSE7b0VZSESYB39mHsgsv0v2hJkKurmc1joXZZrDpuKkQATu/yzBE42avD1
nMUWLAzwwRO5hgrjVTazs1IcZw9Btcx+Q1/uIwyd8XSHjasss8+qK7B1FP0V9Ug
he1G0Uvh2QDbcCjydcXQ==
-----END PRIVATE KEY-----
cloud-id: b1g9shupin2k9j5gomg
folder-id: b1g6gpp7fmeo0hlca4ue
```

Создание бакета и политики доступа для него

Созданный бакет:

```
PS C:\Users\maxpo> yc storage bucket list
```

NAME	FOLDER ID	MAX SIZE	DEFAULT STORAGE CLASS	CREATED AT
devops-course-bucket	b1g6gpp7fmeo0hlca4ue	1073741824	STANDARD	2026-08-27 16:34:59

```
PS C:\Users\maxpo> yc iam mbound
```

Политика доступа внутри бакета:

```
PS C:\Users\maxpo> yc storage bucket get devops-course-bucket --full
name: devops-course-bucket
folder_id: b1g6gpp7fmeo0hlca4ue
anonymous_access_flags:
  read: false
  list: false
  config_read: false
default_storage_class: STANDARD
versioning: VERSIONING_DISABLED
max_size: "1073741824"
policy:
  Statement:
    - Action: s3:ListBucket
      Effect: Allow
      Principal: '*'
      Resource: arn:aws:s3:::devops-course-bucket/*
      Sid: ListLabBucket
    - Action:
      - s3:GetObject
      - s3:PutObject
      - s3:DeleteObject
      Effect: Allow
      Principal: '*'
      Resource: arn:aws:s3:::devops-course-bucket/*
      Sid: ObjectRW
  Version: "2012-10-17"
acl: {}
created_at: "2026-08-27T16:34:59.294274Z"
website_settings:
  redirect_all_requests: {}
resource_id: e3ejjpcitcchm2tgpctk
```

Сохранение файла в бакет:

```
PS C:\Users\maxpo\devops-tms\homework34> yc storage s3api put-object --body ./lab-test.txt --bucket devops-course-bucket --key lab-test.txt
etag: '"b6071b899992ee476aa7c10692b26857"'
request_id: d4e1d34a361fb361
```

Файл в консоли:

☐	Имя	Размер	Класс хранилища	Последнее изменение	Метки	
☐	lab-test.txt	56 Б	Стандартное	27.08.2026, в 20:22	—	...

Получение файла:

```
PS C:\Users\maxpo\devops-tms\homework34> yc storage s3api get-object --bucket devops-course-bucket --key lab-test.txt read.txt
etag: '"b6071b899992ee476aa7c10692b26857"'
request_id: 3d16fb18ff083ad9
accept_ranges: bytes
content_length: "56"
content_type: application/octet-stream
last_modified_at: "2026-08-27T17:22:59Z"

PS C:\Users\maxpo\devops-tms\homework34> cat read.txt
hello from devops-lab-cli
```

Чтение бакета:

```
PS C:\Users\maxpo\devops-tms\homework34> yc storage s3api list-objects --bucket devops-course-bucket
contents:
- key: lab-test.txt
  last_modified: "2026-08-27T17:22:59.326Z"
  etag: '"b6071b899992ee476aa7c10692b26857"'
  size: "56"
  owner:
    id: aje9580lvmpoq70up8q
    display_name: aje9580lvmpoq70up8q
  storage_class: STANDARD
name: devops-course-bucket
max_keys: "1000"
key_count: "1"
request_id: 10dc6c788b937bb4
```

ReadOnly Policy:

Правило 4

...

^

Идентификатор правила

ObjectRO

Результат

Разрешить

Запретить

Принцип выбора

Включить пользовател...

Исключить пользовател...

Пользователь

Все пользователи

Выбрать пользователей

Действие *

GetObject, ListBucket

^v

☐ Все действия

Ресурс

devops-course-bucket/*

devops-course-bucket

Добавить ресурс

Добавить условие

Проверка работоспособности политики:

```
PS C:\Users\maxpo\devops-tms\homework34> yc storage s3api put-object --body ./lab-test.txt --bucket devops-course-bucket --key lab-test.txt
ERROR: operation error S3: PutObject, https response error StatusCode: 403, RequestID: 384dd7d4460b1d33, HostID: , api error AccessDenied: Access Denied

client-trace-id: a601ce20-167a-47aa-ba16-9ddc26f7cdf6

Use client-trace-id for investigation of issues in cloud support
If you are going to ask for help of cloud support, please send the following trace file: C:\Users\maxpo\.config\yandex-cloud\logs\2026-08-27T21-37-17.669-yc
_storage_s3api_put-object.txt
```

Получить можно:

```
PS C:\Users\maxpo\devops-tms\homework34> yc storage s3api get-object --bucket devops-course-bucket --key lab-test.txt read.txt
etag: '"b6071b899992ee476aa7c10692b26857"'
request_id: 563a7d14605a901f
accept_ranges: bytes
content_length: "56"
content_type: application/octet-stream
last_modified_at: "2026-08-27T18:36:37Z"
```

Создание Assume Role и наделение правами на использование S3 бакета через неё

В Yandex Cloud нет Assume Role, но есть механизм имперсонации сервисных аккаунтов. Создаю аккаунт:

```
PS C:\Users\maxpo> yc iam service-account create --name assume-role-account
done (1s)
id: ajel8lbmn0qhtfm63rro
folder_id: blg6gpp7fmeo0hlca4ue
created_at: "2026-09-06T11:31:04Z"
name: assume-role-account
status: ACTIVE

There is a new yc version '1.32.0' available. Current version: '1.29.0'.
See release notes at https://yandex.cloud/ru/docs/cli/release-notes
You can install it by running the following command in your shell:
$ yc components update
```

Права на просмотр папки для сервисного аккаунта:

```
PS C:\Users\maxpo> yc resource-manager folder add-access-binding blg6gpp7fmeo0hlca4ue --role viewer --subject serviceAccount:ajel8lbmn0qhtfm63rro
done (2s)
effective_deltas:
- action: ADD
  access_binding:
    role_id: viewer
    subject:
      id: ajel8lbmn0qhtfm63rro
      type: serviceAccount
```

Имперсонация для сервисного аккаунта:

```
PS C:\Users\maxpo> yc iam service-account add-access-binding ajel8lbmn0qhtfm63rro --role iam.serviceAccounts.tokenCreator --subject userAccount:aje9580lvmpoq70up8q
done (2s)
effective_deltas:
- action: ADD
  access_binding:
    role_id: iam.serviceAccounts.tokenCreator
    subject:
      id: aje9580lvmpoq70up8q
      type: serviceAccount
```

Проверка работы имперсонации:

```
PS C:\Users\maxpo> yc storage s3api list-objects --bucket devops-course-bucket --impersonate-service-account-id ajel8lbmn0qhtfm63rro
contents:
- key: lab-test.txt
  last_modified: "2026-08-27T18:36:37.410Z"
  etag: '"b6071b899992ee476aa7c10692b26857"'
  size: "56"
  owner:
    id: aje9580lvmpoq70up8q
    display_name: aje9580lvmpoq70up8q
  storage_class: STANDARD
name: devops-course-bucket
max_keys: "1000"
key_count: "1"
request_id: 629afa24dbe77b31
```

Создание EC2 и подключение к ней через CLI

Созданная VM:

Имя ↑↓	Статус ↑↓	ОС	Публичный IPv4	Идентификатор ↑↓	RAM	vCPU	Размер дисков	Метки	Cloud Backup	Платформа ↑↓	Доля vCPU	Прерываемая	Зона д	
devops-course-ec2	Running		158.160.135.93	fv4eu3vue691tr2e4cgh	2 ГБ	2	10 ГБ	—	 Не подключён	Intel Ice Lake	100%	Нет	ru-сеп	☆ ...

Ресурсы

Платформа ⓘ Intel Ice Lake

Гарантированная доля vCPU ⓘ 100%

vCPU 2

RAM 2 ГБ

Объём дискового пространства 10 ГБ

Приложение Marketplace

 Ubuntu 24.04 LTS

Тип тарификации Free

Инструкции по настройке

Подключение:

```
PS C:\Users\maxpo> ssh -i ~/.ssh/ssh-key-1788696639207 devops-login-ec2@158.160.135.93
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep  6 12:18:29 UTC 2026

System load:  0.16           Processes:           102
Usage of /:   23.1% of 9.04GB Users logged in:        0
Memory usage: 9%            IPv4 address for eth0: 10.130.0.12
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

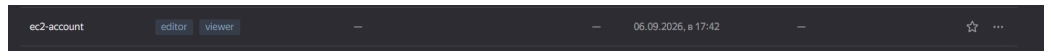
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

devops-login-ec2@devops-course-ec2:~$ |
```

Настройка прав EC2 для работы с S3

В Yandex Cloud нет понятия instance profile, поэтому используется сервисный аккаунт с доступом к S3.

Созданный сервисный аккаунт:



Назначение прав аккаунту на доступ к S3:

```
PS C:\Users\maxpo> yc storage bucket update devops-course-bucket --grants grant-type=grant-type-account,grantee-id=ajehj0ajd0nmcdiq82ua,permission=permission-full-control
name: devops-course-bucket
folder_id: b1g6gpp7fmeo0hlca4ue
default_storage_class: STANDARD
versioning: VERSIONING_DISABLED
max_size: "1073741824"
policy:
  Id: devops-base-policy
  Statements:
    - Action: '*'
      Condition:
        Bool:
          aws:PrincipalIsAWSService: "true"
      Effect: Allow
      Principal: '*'
      Resource:
        - arn:aws:s3:::devops-course-bucket/*
        - arn:aws:s3:::devops-course-bucket
      Sid: AllowAWSServices
    - Action:
      - s3:GetObject
      - s3:ListBucket
      Effect: Allow
      Principal: '*'
      Resource:
        - arn:aws:s3:::devops-course-bucket/*
        - arn:aws:s3:::devops-course-bucket
      Sid: ObjectRO
    - Action:
      - s3:GetObject
      - s3:PutObject
      - s3:DeleteObject
      Effect: Allow
      Principal: '*'
      Resource: arn:aws:s3:::devops-course-bucket/*
      Sid: ObjectRW
    - Action: s3:ListBucket
      Effect: Allow
      Principal: '*'
      Resource: arn:aws:s3:::devops-course-bucket
      Sid: ListABucket
      Version: "2012-10-17"
acl:
  grants:
    - permission: PERMISSION_FULL_CONTROL
      grant_type: GRANT_TYPE_ACCOUNT
      grantee_id: ajehj0ajd0nmcdiq82ua
  created_at: "2026-08-27T16:34:59.294274Z"
  resource_id: e3ejjpcitcche2t0pctk
```

Привязка аккаунта к EC2:

```
PS C:\Users\maxpo> yc compute instance update fv4eu3vue69ltr2e4cgh --service-account-id ajehj0ajd0nmcdiq82ua
done (8s)
id: fv4eu3vue69ltr2e4cgh
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-06T12:11:01Z"
name: devops-course-ec2
zone_id: ru-central1-d
platform_id: standard-v3
resources:
  memory: "2147483648"
  cores: "2"
  core_fraction: "100"
status: RUNNING
boot_disk:
  mode: READ_WRITE
  device_name: fv45orn2gltl0o2gfho9
  auto_delete: true
  disk_id: fv45orn2gltl0o2gfho9
network_interfaces:
  - index: "0"
    mac_address: d0:0d:ef:0f:fe:71
    subnet_id: fl04u3dejsaa66evvufp
    primary_v4_address:
      address: 10.130.0.12
      one_to_one_nat:
        address: 158.160.135.93
        ip_version: IPV4
fqdn: devops-course-ec2.ru-central1.internal
scheduling_policy: {}
service_account_id: ajehj0ajd0nmcdiq82ua
network_settings:
  type: STANDARD
placement_policy: {}
gpu_settings: {}
metadata_options:
  gce_http_endpoint: ENABLED
  aws_v1_http_endpoint: ENABLED
  gce_http_token: ENABLED
  aws_v1_http_token: DISABLED
  aws_v2_http_endpoint: ENABLED
  aws_v2_http_token: ENABLED
serial_port_settings:
  ssh_authorization: OS_LOGIN
hardware_generation:
  legacy_features:
    pci_topology: PCI_TOPOLOGY_V2
application: {}
```

Проверка доступности к S3 из EC2:

```
devops-login-ec2@devops-course-ec2:~$ yc storage s3api get-object --bucket devops-course-bucket --key lab-test.txt ./downloaded-lab-test.txt
etag: '"b6071b89992ee476aa7c10692b26857"'
request_id: 6fcac5a94cd81a1c
accept_ranges: bytes
content_length: "56"
content_type: application/octet-stream
last_modified_at: "2026-08-27T18:36:37Z"

devops-login-ec2@devops-course-ec2:~$ cat downloaded-lab-test.txt
❖hello from devops-lab-cli

devops-login-ec2@devops-course-ec2:~$ cat lab-test-ec2.txt
hello from ec2

devops-login-ec2@devops-course-ec2:~$ yc storage s3api put-object --bucket devops-course-bucket --key lab-test-ec2.txt
etag: '"d41d8cd98f00b204e9800998ecf8427e"'
request_id: a40c5ec4c8a489fa
```

Объекты						
Поиск по началу имени объекта, включая префикс ⓘ						
☐	Имя	Размер	Класс хранилища	Последнее изменение	Метки	⚙
☐	 lab-test-ec2.txt	0 Б	Стандартное	06.09.2026, в 18:34	—	...
☐	 lab-test.txt	56 Б	Стандартное	27.08.2026, в 21:36	—	...

Скрипт для удаления всех сущностей внутри репозитория

Работа скрипта:

```
gabka@DESKTOP-08RCOSC:/mnt/c/Users/maxpo/devops-tms/homework34$ ./cleanup-yc-config.sh
=== Очистка ресурсов Yandex Cloud ===
Продолжить? [y/N] y
Удаление объектов из бакетов...
Очистка бакета: devops-course-bucket
delete: s3://devops-course-bucket/lab-test-ec2.txt
delete: s3://devops-course-bucket/lab-test.txt
done (11s)
Удаление виртуальных машин...
Удаление VM: fv4eu3vue69ltr2e4cgh
id: fv4pcup30du5rplnht8a
description: Delete instance
created_at: "2026-09-06T16:43:41.883923210Z"
created_by: ajehjgj73jidp7hjquio
modified_at: "2026-09-06T16:43:41.883923210Z"
metadata:
  '@type': type.googleapis.com/yandex.cloud.compute.v1.DeleteInstanceMetadata
  instance_id: fv4eu3vue69ltr2e4cgh
Удаление сервисных аккаунтов...
Удаление SA: aje9580lvmpoq70up8q
done (8s)
Удаление SA: ajec5j44j2j24m7oev0l
done (12s)
Удаление SA: ajehj0ajd0nmcdiq82ua
ERROR: rpc error: code = FailedPrecondition desc = The deleting entity is blocked by references:
Referrer type | Reference type | Resource ID
compute.instance | USED_BY | fv4eu3vue69ltr2e4cgh
gabka@DESKTOP-08RCOSC:/mnt/c/Users/maxpo/devops-tms/homework34$ ./cleanup-yc-config.sh
=== Очистка ресурсов Yandex Cloud ===
Продолжить? [y/N] y
Удаление объектов из бакетов...
Удаление виртуальных машин...
Удаление сервисных аккаунтов...
Удаление SA: ajehj0ajd0nmcdiq82ua
done (8s)
Удаление SA: ajel8lbmn0qhtfm63rro
done (8s)
✅ Очистка завершена!
```